1. Meet the policy and open the data

Question: Did North Carolina's 2013 UI cuts reduce unemployment?

Name: _____ Date: _____

In February 2013, North Carolina passed legislation reducing UI generosity. The changes took effect in July 2013. In this exercise, we will study how to quantify the effect using difference-in-differences.

Sequence: start with a two-state table and simple regression, inspect monthly trends, and then expand the time window and comparison group.

Step 1A. Extract the package. Save a new do-file as `my_nc_lab.do`. Replace the example path below with the extracted folder's path. Use forward slashes on Windows too. Run complete blocks from the Do-file Editor.

```
clear all
set more off
cd "/YOUR/PATH/NC_UI_DiD_Teaching_Package"
capture log close
log using "my_nc_lab.log", text replace
import delimited "nc_ui_monthly_2010_2018.csv", clear
describe
list state year month unemployment_rate in 1/8
```

Step 1B. Declare and check the monthly panel.

```
gen mdate = ym(year,month)
format mdate %tm
isid state_id mdate
xtset state_id mdate
count
tab state
summarize unemployment_rate
```

Record the unit of observation, number of observations, and number of states.

Check: 2,244 observations: 22 states over 102 months, July 2010–December 2018. An unemployment rate of 8 means 8%, not 0.08%.

Why would a before/after comparison of NC alone be insufficient?

1C. Construct the variables for DiD

The CSV contains only state identifiers, calendar time, and the unemployment rate. You must construct the treatment indicators yourself. Use the same notation as the lecture:

$$Y_{i,t} = \alpha + \beta NC_i + \lambda After_t + \tau(NC_i \times After_t) + \varepsilon_{i,t}.$$

Policy implementation: July 2013. For this exercise, every month before July 2013 is coded as before; July 2013 onward is coded as after. The after indicator must be defined for *every state*, not just NC.

1. Create `NC`: one for North Carolina and zero for every other state. Is this variable allowed to change across months within a state?

2. Create `after`: one in July 2013 and all later months, and zero before July 2013. Explain why `month >= 7` alone would be incorrect in a multi-year dataset.

3. Create `did`: the product of `NC` and `after`. For which state-month observations is it one?

Complete these commands, replacing each blank before running. Stata logical comparisons return one when true and zero when false. Use `mdate` and `ym(2013,7)` for the implementation cutoff.

```
gen NC = ______________________________
gen after = ___________________________
gen did = ______________________________
```

Check your coding:

```
tab state NC
tab NC after
list state mdate NC after did ///
    if inlist(state,"NC","VA") & year==2013, sepby(state)
```

In the NC rows, `after` and `did` should switch from zero to one in July 2013. In VA, `after` also switches in July, but `did` stays zero. All three indicators should have no missing values.

Complete the definitions: In the slides, NC_i corresponds to Stata's `NC`; $After_t$ corresponds to `after`; $NC_i \times After_t$ corresponds to `did`.

1D. Start simple: North Carolina and Virginia

Use North Carolina as the treated state and Virginia as the neighboring comparison state. The outcome is the seasonally adjusted **unemployment rate**, not benefit receipt or benefit exhaustion.

Before: July–December 2012. **After:** July–December 2013. We compare the same six calendar months in each year. The after window ends before the nationwide federal emergency-benefit expiration at the end of December 2013. Virginia is an illustrative comparison, not an automatically valid counterfactual.

Temporarily restrict the sample:

```
preserve
keep if inlist(state,"NC","VA")
keep if inlist(year,2012,2013) & inrange(month,7,12)
assert NC == (state=="NC")
count
tab state after
bysort state after: summarize unemployment_rate
```

Check: 24 observations, six in each state-period cell. `NC` equals one for NC; `after` equals one for the after period.

Use the means from Stata to complete the table. Keep at least three decimal places until the final answer. Differences are in percentage points.

	Before	After	After − Before
	Jul–Dec 2012	Jul–Dec 2013	
North Carolina			
Virginia			
NC − VA			

$$\widehat{\tau} = (\bar{Y}_{NC,after} - \bar{Y}_{NC,before}) - (\bar{Y}_{VA,after} - \bar{Y}_{VA,before}).$$

My DiD estimate: _____ percentage points.

What is NC's predicted after-period unemployment rate without the reform, if NC had changed by the same amount as VA? Why is NC's decline alone not the DiD estimate?

Keep this temporary sample for the next page. Do not restore yet.

1E. Estimate the same comparison in Stata

Estimate the lecture equation using the same 24 state-month observations:

$$Y_{i,t} = \alpha + \beta NC_i + \lambda After_t + \tau(NC_i \times After_t) + \varepsilon_{i,t}.$$

Here NC_i is `NC`, and $After_t$ is `after`.

```
reg unemployment_rate i.NC##i.after
scalar simple_did = _b[1.NC#1.after]
display "NC-VA DiD (percentage points): " simple_did
```

The `##` syntax includes both main effects and their interaction.

Coefficient	Meaning	Estimate
`_cons`	VA before	
`1.NC`	NC − VA before	
`1.after`	VA's change	
`1.NC#1.after`	NC's change − VA's change	

1. Does the interaction coefficient equal your table calculation? Why?
2. Interpret its sign and magnitude in one sentence.
3. Why is the after-period coefficient alone not the policy effect?
4. State the parallel-trends assumption for NC and VA. What could violate it?
5. Does lower unemployment necessarily mean more people found jobs?

Focus on point estimates. Default regression standard errors are not appropriate for inference with dependent state-month observations. With two states and only North Carolina receiving the reform, conventional state-clustered significance tests are not a credible solution either. This exercise demonstrates the equality of the table calculation and interaction coefficient.

Restore the full data before continuing:

```
restore
count
```

Check: 2,244 observations. The next sections expand to all 21 comparison states and July 2010–December 2013. Use the same `after` variable there. The estimate may differ because both the comparison group and before window change.

2. Plot the raw trends

A two-period table hides the monthly paths. We now inspect those paths and broaden the comparison group. For the main exercise, stop in December 2013: federal emergency benefits expired nationally at the end of that year, changing the contrast afterward.

```
keep if mdate <= ym(2013,12)
tab NC after
list state mdate NC after did ///
    if state=="NC" & year==2013, noobs
preserve
collapse (mean) unemployment_rate, by(mdate NC)
local implement = ym(2013,7)
twoway ///
 (line unemployment_rate mdate if NC==1, ///
     lcolor(navy) lwidth(medthick)) ///
 (line unemployment_rate mdate if NC==0, ///
     lcolor(maroon) lwidth(medthick)), ///
 xline(‘implement’, lpattern(dash)) ///
 legend(order(1 "North Carolina" 2 "Comparison mean")) ///
 ytitle("Unemployment rate (%)") xtitle("Month") ///
 title("Raw unemployment trends")
graph export "my_raw_trends.png", replace width(1800)
restore
```

Why preserve/restore? `collapse` temporarily replaces state-month rows with group-month means. `restore` returns the state-level panel. Comparison states receive equal weights.

In the lecture notation, $NC_i = 1$ for NC and $After_t = 1$ from July 2013 onward. In Stata these are `NC` and `after`; `did` is their product.

1. Do the groups have the same unemployment levels? Must they for DiD?
2. Before July 2013 implementation, do the groups move similarly?
3. Does NC's decline begin before or after July implementation?

3. Align the lines and assess pre-policy trends

Shift the comparison mean by a constant so it matches NC in January 2013, as a visual reference point. This is a graphing choice; the treatment indicator still starts in July.

```
preserve
collapse (mean) unemployment_rate, by(mdate NC)
quietly summarize unemployment_rate ///
    if NC==1 & mdate==ym(2013,1)
local nc_jan = r(mean)
quietly summarize unemployment_rate ///
    if NC==0 & mdate==ym(2013,1)
local shift = `nc_jan' - r(mean)
gen aligned_rate = unemployment_rate + `shift'*(NC==0)
local implement = ym(2013,7)
twoway ///
 (line aligned_rate mdate if NC==1, ///
     lcolor(navy) lwidth(medthick)) ///
 (line aligned_rate mdate if NC==0, ///
     lcolor(maroon) lwidth(medthick)), ///
 xline(`implement', lpattern(dash)) ///
 legend(order(1 "North Carolina" ///
              2 "Comparison mean (shifted)")) ///
 ytitle("Unemployment rate (%)") xtitle("Month") ///
 title("Aligned at January 2013")
graph export "my_aligned_trends.png", replace width(1800)
restore
```

1. Does a constant shift change slopes, monthly changes, or the DiD estimate?
2. Identify a pre-July 2013 period supporting or challenging the comparison.
3. Does NC begin to diverge from the comparison group before July 2013? What does this suggest about the credibility of the comparison?

Parallel trends: without the reform, NC and the comparison group would have experienced the same average changes. Similar pre-policy trends provide evidence, but do not prove this assumption. Matching levels at one date does not validate a counterfactual.

4. Calculate DiD with the broader sample

Use July 2010–June 2013 as before and July–December 2013 as after. Use actual unemployment rates, not shifted graph values.

```
preserve
collapse (mean) unemployment_rate, by(NC after)
list NC after unemployment_rate, noobs sepby(NC)
restore
```

	Before	After	After − Before
North Carolina			
Comparison mean			
NC − comparison			

Check your arithmetic:

```
quietly summarize unemployment_rate if NC==1 & after==0
scalar nc_before = r(mean)
quietly summarize unemployment_rate if NC==1 & after==1
scalar nc_after = r(mean)
quietly summarize unemployment_rate if NC==0 & after==0
scalar c_before = r(mean)
quietly summarize unemployment_rate if NC==0 & after==1
scalar c_after = r(mean)
scalar did_manual = (nc_after-nc_before)-(c_after-c_before)
display "Manual DiD (percentage points): " did_manual
```

My estimate: _____ percentage points.

Relative to the comparison group, NC's unemployment rate ____ by approximately ____ percentage points.

Why might this estimate differ from the NC–VA exercise? Which parts of the design changed? Does the result necessarily mean employment increased? Explain.

5. Estimate DiD using regression

Step 5A. Basic interaction regression:

$$Y_{i,t} = \alpha + \beta NC_i + \lambda After_t + \tau(NC_i \times After_t) + \varepsilon_{i,t}.$$

```
reg unemployment_rate i.NC##i.after, vce(cluster state_id)
scalar did_basic = _b[1.NC#1.after]
display "Basic regression DiD: " did_basic
```

Step 5B. State and calendar-month fixed effects:

$$Y_{i,t} = \alpha_i + \sum_{j \neq t_0} \gamma_j \mathbf{1}\{t = j\} + \tau(NC_i \times After_t) + \varepsilon_{i,t}.$$

Each state has its own baseline α_i. Each year-month has a common effect relative to omitted month t_0; calendar-month effects are not simply month-of-year effects.

```
xtreg unemployment_rate did i.mdate, fe vce(cluster state_id)
estimates store baseline
scalar did_fe = _b[did]
display "Fixed-effects DiD: " did_fe
display "Difference from manual: " did_fe-did_manual
```

Method	Estimate (percentage points)
Manual calculation	
Basic interaction regression	
State/month fixed effects	

1. Why do these point estimates agree in this balanced, equally weighted panel?
2. What do state effects and year-month effects account for?
3. Why is the interaction, rather than the after indicator alone, of interest?

Inference note: state clustering allows within-state dependence, but NC is the only state receiving the reform. Conventional clustered p-values may be unreliable. Focus on point estimates and identification assumptions; significance alone cannot establish causality.

6. Interpret the implementation-based estimate

All estimates in this exercise use **July 2013 implementation** as the treatment cutoff. We do not estimate a separate effect for passage or remove a separate timing interval.

Record your preferred full-sample DiD estimate: _____ percentage points.

Discuss:

1. Interpret the sign and magnitude. Is this NC's overall unemployment change, or NC's change relative to the comparison group?

2. How does the broader-sample estimate compare with the initial NC–VA estimate? Why might they differ?

3. What do the monthly graphs suggest about the equal-change assumption before July 2013? Does a similar pre-period pattern prove parallel trends after implementation?

Policy memo: 150–200 words.

1. State the implementation-based estimate, its sign, and its units.
2. Explain what the graphs suggest about the choice of comparison group.
3. Discuss two threats to causality, such as different recovery paths, industry shocks, or other state policies.
4. Explain why unemployment alone cannot establish employment or welfare gains. Name an additional outcome you would examine.

```
log close
```

Submit: do-file and log; your definitions of `NC`, `after`, and `did`; raw and aligned graphs; completed tables; and memo. A careful qualified conclusion earns credit even if you find the design unconvincing.

Optional extension: plot through 2018. Explain why nationwide federal emergency-benefit expiration at the end of 2013 complicates a longer after-policy comparison. Retain July 2013 as the implementation cutoff.

Data and scope. Revised BLS LAUS monthly data supplied in the package. This teaching adaptation follows the aggregate application in Dahl and Knepper, *Unemployment Insurance, Starting Salaries, and Jobs*, NBER WP 30152, Figure 2. The broader exercise uses 21 regional comparison states; the supplied notes report that the paper uses 20. It is not an exact replication. See README for the state list and provenance. NC's reform bundles benefit amount, duration, and federal benefit changes; this exercise does not identify their separate effects.

`https://www.nber.org/papers/w30152`
`https://download.bls.gov/pub/time.series/la/`